

Civics Group	Index Number	Name (use BLOCK LETTERS)
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H2



ST. ANDREW'S JUNIOR COLLEGE
2014 JC2 Preliminary Examinations

H2 BIOLOGY

9648/2

Paper 2: Core (Mark Scheme)

Tuesday

2nd September 2014

2 hours

Additional Materials: Answer Paper
Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A (Structured Questions)

Answer **all seven** questions.

Write your answers in the spaces provided on the question paper.

Section B (Essay Question)

Answer **one** essay question.

Write your answers on the separate answer paper provided.

All working for numerical answers must be shown.

INFORMATION TO CANDIDATES

At the end of the examination,

1. Attach Section B answers to the **cover sheet** provided.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/10
2	/12
3	/10
4	/12
5	/13
6	/11
7	/12
Total	/80

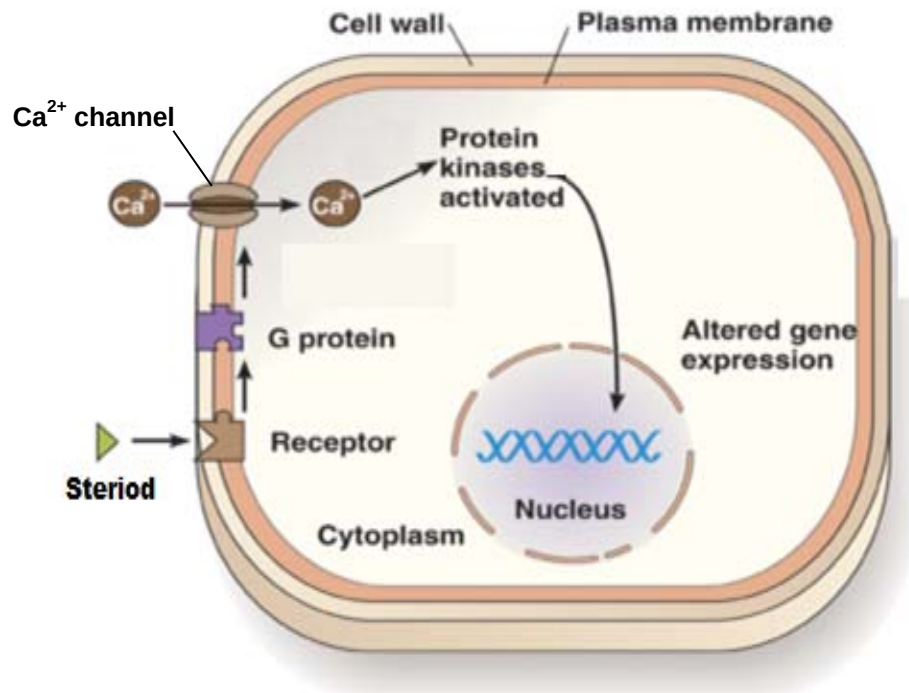
This document consists of **xx** printed pages.

[Turn over

Section A

Answer **all** questions.

1(a) Fig. 1.1 shows the plant cell transmembrane cell signalling mechanism which is comparatively less well-studied than that of the animal cell.



Source: Adapted from <https://plantsrock.wikispaces.com/file/view/Picture1.png/30513514/409x329/Picture1.png>

Fig 1.1

(i) State the main stages of *cell signalling*.

.....[1]

- 1 Ligand-receptor interaction
/ Signal (molecule preferred) detected at the cell's surface;
 - 2 Signal transduction
/ transmitted into the cell's interior;
 - 3 Cellular response
/ changes in gene expression / cell behaviour;
- [Reward 1 mark for all 3 points present]

- (ii) With reference to **Fig. 1.1**, describe the signal transduction within the plant cell upon steroid binding.

-[4]
- 1 (Binding of steroid to binding site on extracellular side of receptor triggers) conformational changes which activates the receptor ;
 - 2 exposes the cytoplasmic (G protein binding) domain of the receptor
 - 3 G-protein binds and exchanges GDP for GTP;
 - 4 G-protein is activated;
 - 5 G-protein dissociates from the receptor, then diffuses along the cell membrane ;
 - 6 (Binds to and) open the Ca^{2+} channel which allows the influx of Ca^{2+} ions (into the cell);
 - 7 Ca^{2+} ions acts as a second messenger
 - 8 by activating protein kinases;
 - 9 which activate proteins via phosphorylation;
 - 10 ref. signal amplification / e.g. each Ca^{2+} ion activate a large number of protein kinase thus amplifying the signal;

According to Michael Hothorn of the Salk Institute, the transmembrane signalling mechanism used by plants differ from those used by animals. A small steroid molecule on the outside of the plant cell assists in the transmembrane signalling process. By contrast, this sort of molecule and its receptor is generally located intracellularly in animal cells.

- (iii) Explain how such signal molecules enter the cytoplasm of the animal cell.

-[1]
- 1 ref. signal molecule is non-polar/hydrophobic
 - 2 able to pass through the hydrophobic core (of the cell surface membrane & nuclear envelope)
 - 3 consisting of fatty acid tail of the phospholipid bilayer
 - 4 ref. diffusion

Beta cells in the Islets of Langerhans in the pancreas secrete peptide hormone, insulin into the blood stream. The production of functional insulin and its secretion is done with the help of vesicles.

Proteins embedded in the membranes of vesicles have different functions.

- COPII and v-SNARE proteins are known as 'address proteins'.
- Vesicles that transport insulin from the rough endoplasmic reticulum to the Golgi body are coated with COPII proteins.
- Vesicles that transport insulin from the Golgi body to the cell surface membrane are coated with v-SNARE proteins.

(b) Outline how the vesicles are moved from one cellular location to another and suggest how these 'address proteins' ensure that vesicles are transported to the correct locations within the cell.

.....[2]
How vesicles move from one organelle to another [compulsory 1 mark]

- 1 (vesicle) moves along cytoskeleton / microtubule / microfilament
- 2 which provide pathways / tracks (for movement)
- 3 Uses ATP as energy source

How address proteins ensure correct cell destinations [compulsory 1 mark]

- 4 Specific receptor found only on target organelle / cell surface membrane
- 5 COPII / v-SNARE / address protein has a **specific** 3D conformation
- 6 Conformation of receptor and 'address protein' are complementary
 [ACCEPT]: shape of protein fits the receptor

Insulin travels via the bloodstream to bind to the insulin receptors embedded in the plasma membrane on target cells. The plasma membrane can be described by the fluid mosaic model.

(c) Describe what is meant by the fluid mosaic model.

.....[2]

'Fluid' refers to

- 1 the phospholipids and proteins being free to move **within** the membrane, and
- 2 the phospholipids also being able to move **across** the layers/transversely ;

'Mosaic' refers to the

- 3 proteins being embedded in the phospholipid bilayer ;
- 4 in a scattered / random arrangement;

[Q1 Total: 10]

2(a). A strain of the bacterium, *Streptococcus pneumonia*, has been found which is resistant to the antibiotic vancomycin. This vancomycin resistance is caused by a mutation of a gene coding for an enzyme, S.

(i) Suggest and briefly explain 2 different ways in which a mutation of the gene may still result in the production of a functional enzyme.

.....[2]

- 1 Same amino acid encoded for / silent mutation (due to degeneracy of genetic code);
- 2 Different amino acid encoded for but has similar R group properties (as original amino acid) ;
- 3 Different amino acid encoded for but amino acid is not crucial for the function of protein;
- 4 Ref. Substitution of one base pair for another ;
- 5 Does not alter the overall 3D shape of the enzyme / enzyme active site ;

(ii) Explain briefly how a mutation of the gene may result in an enzyme with reduced activity.

.....[2½]

- 1 Different codon/mRNA sequence encoded for;
- 2 Different amino acid encoded for / different primary structure (of polypeptide) ;
- 3 Different R group (as original amino acid) ;
- 4 Different 3D configuration of polypeptide / enzyme ;
- 5 Different 3D shape of active site ;
- 6 Different affinity of active site for substrate ;
- 7 Different amino acid / stop codon is located near the end of gene (not much effect on the folding of polypeptide) ;

(iii) A random mutation resulting in resistance to vancomycin was found to occur on a plasmid. Explain two ways how the resistance to vancomycin may spread naturally amongst bacteria.

.....[3]

Transformation

- 1 Release of plasmid (containing vancomycin resistance gene) into environment ;
- 2 By dead / lysed bacteria donor cells ;
- 3 Uptake of plasmid ;
By naturally competent bacteria cells ;

Conjugation

- 4 **Contact** between recipient F^- and donor F^+ cells;
- 5 Involve formation of sex pilus ;
- 6 Transfer of vancomycin resistance gene found on F plasmid ;

Transduction

- 7 Vancomycin resistance gene from the host/donor bacterium is transferred to the recipient bacterium cell via bacteriophage
- 8 Donor DNA from the host is accidentally packaged within viral capsid.
- 9 Resultant phages infect other bacteria
- 10 Ref. transformation, conjugation and transduction (any 2 ways)

REJECT Binary fission. The phrase “spread naturally amongst bacteria” implied horizontal gene transfer.

The control of the expression of the lac operon genes, which allow uptake and digestion of lactose in the bacterium *Escherichia coli*, is well known.

Fig.2.1 shows the arrangement of the elements of the *lac* operon.

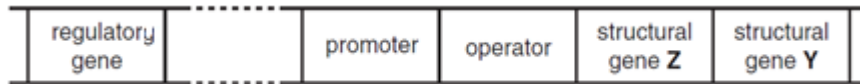


Fig.2.1

(b)(i) Distinguish between structural and regulatory gene.

-[1]
- 1 A structural gene is a region of DNA that codes for a protein that forms part of a structure or has an enzymatic function ;
 - 2 A regulatory gene codes for a specific protein product that regulates the expression of the structural genes;

(ii) Describe how genes Z and Y are switched on in bacteria that are moved to a nutrient medium that contains lactose.

-[3½]
- 1 Lactose enters the bacteria via its qualified leaky membrane ;
 - 2 Lactose isomerize to form allolactose / ref. allolactose as an inducer;
 - 3 Lac repressor constitutively synthesized in active form ;
 - 4 Allolactose binds to the lac repressor at its allosteric site, changing its conformation at the DNA-binding site ;
 - 5 Lac repressor is inactive and no longer binds to the operator ;
 - 6 Operator site is exposed;
 - 7 RNA polymerase has access to structural genes / RNA polymerase transcribes the structural genes ;

[Q2 Total: 12]

3. The following is an excerpt taken from the Health Promotion Board's website .

"Influenza is a respiratory illness which is highly contagious. It is a serious condition because the infection, at times, can lead to complications and even death. The best way to prevent the flu is by getting a flu vaccination each year."

(a) Describe the structural features of the influenza genome.

.....[1½]

- 1 Single-stranded RNA;
- 2 Negative-sense RNA;
- 3 8 segments;
- 4 Each RNA segment associated with capsid proteins;

(b) Suggest why the influenza vaccine has to be changed each year.

.....[3½]

- 1 Correct ref. to antigenic drift and antigenic shift ;
- 2 virus is able to reassort RNA segments
- 3 Occurs when two different strains of virus infect the same cell
- 4 spontaneous mutation in RNA genome
/ virus' genes are made up of RNA, it can mutate more frequently
- 5 due to lack of proof-reading in RNA-dependent RNA polymerase
- 6 Protein / **antigen** (haemagglutinin and neuraminidase) encoded is changed ;
- 7 Cannot be detected by the antibodies present in the immune system ;
- 8 Difficult to provide one influenza vaccination that will protect an individual for life;

Tamiflu[®] is an antiviral drug that can be used to treat influenza patients.

(c) State why a doctor would not prescribe antibiotics to treat influenza.

.....[½]

- 1 Antibiotics are not effective against viruses ;
REJECT viruses are resistant to antibiotics
 / Only effective against bacteria (and fungi / protozoa).
ACCEPT correct reference to detail of antibiotic action e.g. 'antibiotics attack cell wall which is not present in influenza virus'.

Tamiflu[®] is a neuraminidase inhibitor. **Fig.3.1** shows the structures of Tamiflu[®] and sialic acid.

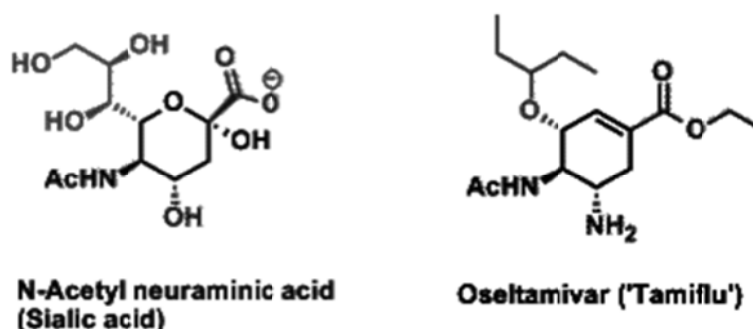


Fig.3.1

(d) With reference to **Fig.3.1**, explain how Tamiflu[®] could be administered to inhibit neuraminidase.

.....[2½]

- 1 Tamiflu[®] has similar structure to sialic acid ;
- 2 Competitive inhibitor of neuraminidase enzyme ;
- 3 Competes with sialic acid substrate for the active site of neuraminidase ;
- 4 Decrease in frequency of effective collisions between neuraminidase and sialic acid;
- 5 Decrease in enzyme-substrate complexes produced;

(e) Suggest how Tamiflu[®] could help to reduce the spread of influenza.

.....[2]

- 1 Prevent enzymatic removal of sialic acid on the viral envelope derived from host membrane
- 2 Causing agglutination of the enveloped virus,;
- 3 thus preventing new viruses from being released
 / Viruses cannot bud off from infected cells ;
- 4 Cannot infect / spread to other cells.

[Q3 Total: 10]

4. Neuregulin 1 (NRG1) gene is a potential gene which contributes to the development of breast cancers. Results of a study (Chua *et al.*, 2009) on DNA methylation and gene expression of the NRG1 gene in different breast cell lines were shown below in Fig. 4.1 and 4.2.

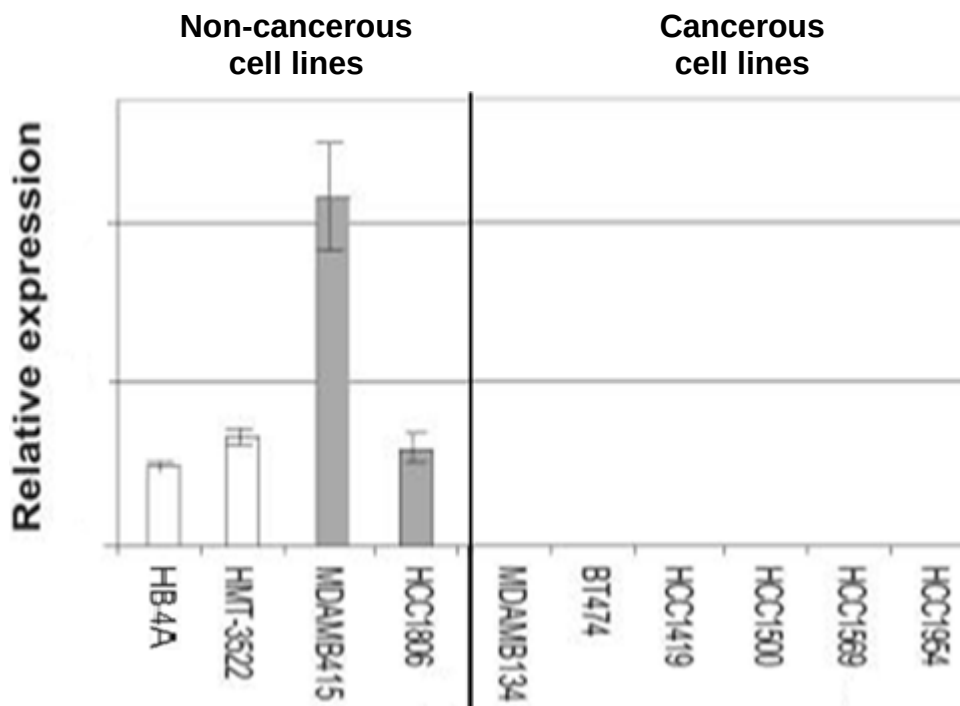


Fig. 4.1. Relative expression of NRG1 gene in various cell lines. Names of cell lines are reflected on the x-axis.

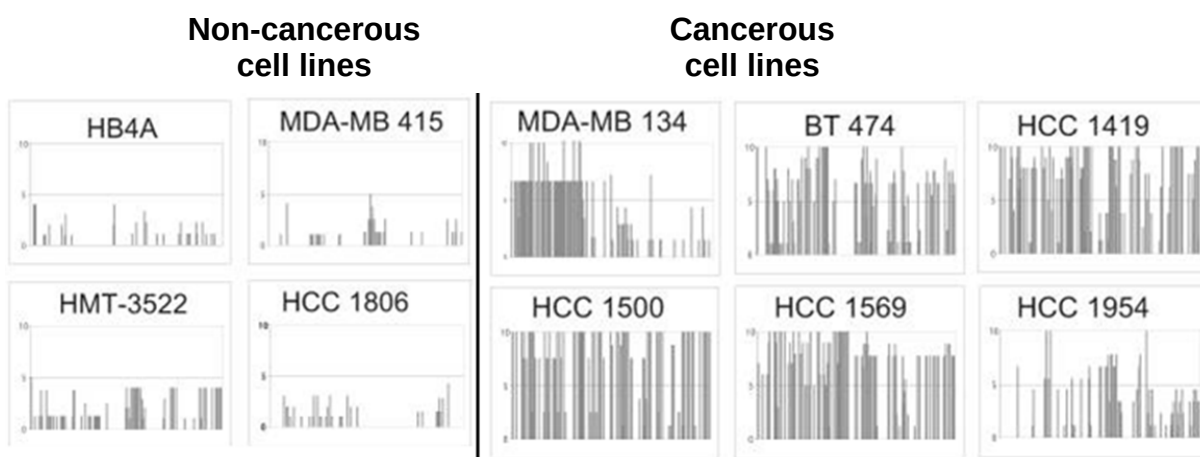


Fig. 4.2. Extent of DNA methylation of NRG1 gene, as represented by the quantity of vertical lines present, in the various cell lines.

(a) With reference to **Fig. 4.1** and **4.2**, explain the relationship between the extent of DNA methylation and NRG1 gene expression.

.....[2½]

Relationship:

- 1 The higher the extent of DNA methylation, the lower the NRG1 gene expression / high levels of DNA methylation silences gene expression;
- 2 Gene is expressed in non-cancerous cell lines/HB4A/MDA-MB415/HMT-3522/HCC1806 cell lines when there is less DNA methylation;
- 3 Gene is silenced / not expressed in cancerous cell lines/ MDA-MB134/BT474/HCC1419/HCC1500/HCC1569/HCC1954 cell lines when there is extensive DNA methylation;

Explain:

- 4 ref. RNA polymerase cannot recognize promoter;
- 5 due to different 3D shape of recognition site after DNA methylation;
- 6 ref. recruitment of histone deacetylases;
- 7 histone deacetylases remove acetyl group from lysine residues of histone tails;
- 8 ionic attraction/interaction between positive histone tail and negative DNA can form again;
- 9 makes the DNA more compact / ref. heterochromatin;
(therefore, reduces transcription of genes)

(b) The expression of some other genes may be regulated negatively at the post-translational level. List two examples of such regulation.

.....[1]

- 1 Phosphorylation of proteins (to decrease its function);
(Regulate cellular activity / influence biological activity)
- 2 Ubiquitination marks protein for degradation/ ref to ubiquitin & proteasome;
(allows control of protein activities
/ prevent aberrant activities so that proteins will not stay too long in cytoplasm)

(c) Cancer is a multi-step process involving the mutations of both proto-oncogenes and tumour suppressor genes. With reference to gene amplification, explain how proto-oncogenes can be converted to oncogenes.

.....[1½]

- 1 Error in cytokinesis / unequal crossing over / DNA slippage (leads to gene amplification);
- 2 generates multiple copies of the proto-oncogene;
- 3 ref. to gene is overexpressed, forming excessive amount of protein; [Reject: production of abnormal protein]

(d) Cytoplasmic mRNA of all expressed genes was isolated from the BT474 cell line. This RNA was mixed with a purified, denatured DNA fragment that carried a large intron of a housekeeping gene (a gene expressed in essentially all cells), and the mixture was incubated for 12 hours under re-naturation conditions.

Explain whether RNA-DNA duplexes would be formed in this experiment.

.....[1]

- 1 RNA-DNA duplexes would not be formed;
Explain:
- 2 Intron sequences are excised by spliceosomes before exporting out to the cytoplasm
/ splicing would have occurred to remove the introns;

(e) The end-replication problem explains why normal somatic cells are limited in the number of mitotic divisions before they die out.

(i) Outline the end-replication problem in eukaryotic chromosomes.

.....[3]

- 1 DNA polymerase can only add nucleotides to the 3'-OH end
- 2 of a elongating/pre-existing polynucleotide
- 3 primer (bound to the very end of the template strand) removed
- 4 but cannot be replaced with DNA
/ gap cannot be filled in
- 5 because no 3'-OH available (to which a DNA nucleotide can be added)
- 6 the 5' end of the DNA becomes shorter relative to that of the previous generation

(ii) Explain how cells of the BT474 cell line overcome the end-replication problem.

.....[3]

- 1 BT474 is a cancerous cell line;
- 2 Telomerase gene is **activated** in the cancerous cells; [Reject: mutated telomerase gene]
- 3 Telomerase adds telomere repeat sequences to the 3' OH end of template DNA strands;
- 4 via complementary base pairing using RNA as a template;
- 5 DNA polymerase is then able to fill up the gaps at the 5' ends of daughter strands;
- 6 thus, compensating for the shortening that occurs during DNA replication
/ serves as a disposable buffer to prevent gene erosion;

[Q4 Total: 12]

5. In a particular species of corn harvested from a vegetable farm, the allele *A* allows the deposition of anthocyanin (blue) pigment in the kernels (seeds), while *aa* plants have yellow kernels. At a second gene locus, *B*— produces smooth kernels, while *bb* kernels are wrinkled.

A plant, heterozygous for blue smooth kernels was crossed to a plant with yellow wrinkled kernels. The progeny consisted of 1447 blue smooth, 169 blue wrinkled, 186 yellow smooth, and 1510 yellow wrinkled.

(a)(i) Define the term gene locus.

.....[1]

- 1 The fixed position on a particular chromosome;
- 2 that each gene occupies;

(ii) Define the term heterozygous.

.....[1]

- 1 Genotype of two different alleles of a gene;
- 2 at a particular gene locus of homologous chromosomes;

A chi-squared test was carried out for this cross.

<i>Phenotypes</i>	<i>Observed number (O)</i>	<i>Expected number (E)</i>	$\frac{(O - E)^2}{E}$
Blue smooth	1447	828	462.75
Blue wrinkled	169	828	524.49
Yellow smooth	186	828	497.78
yellow wrinkled	1510	828	561.74
$\chi^2 = 2046.78$			

Table 1: A chi-squared table.

degrees of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

(b)(i) Use the calculated value of χ^2 of 2046.78, and the table of probabilities provided in **Table 1** to find the probability of the observed results departing significantly by chance from the expected ratio.

.....[½]

- 1 Probability < 0.001
/ Probability < 0.1%

(ii) State what conclusions may be drawn from the probability found in (b)(i).

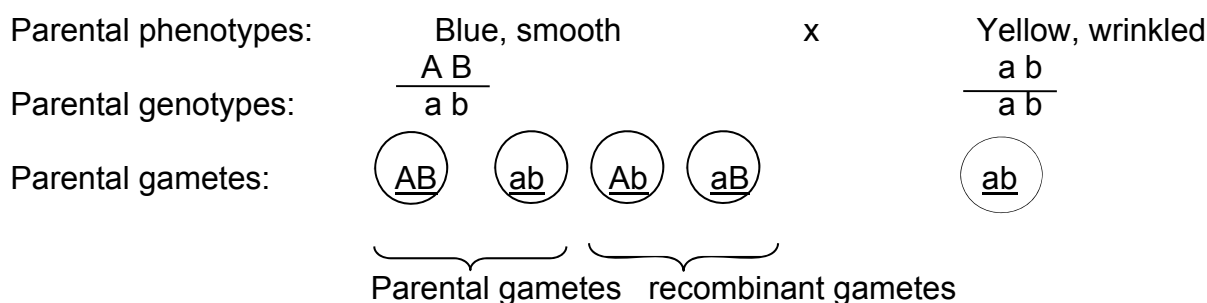
.....[1½]

- 1 there is a significant difference between the observed and the expected values at the **p=0.05 / 5% level** ;
- 2 Any difference is not due to chance;
- 3 predicted phenotype ratio of 1:1:1:1 is incorrect;

[no marks awarded if (i) is wrong]

(c) Using the symbols provided, draw a genetic diagram for the above cross.

.....[4]



Punnett square
showing F₁ genotypes:

	$\frac{AB}{}$	$\frac{ab}{}$	$\frac{Ab}{}$	$\frac{aB}{}$
$\frac{ab}{}$	$\frac{AB}{ab}$ □	$\frac{ab}{ab}$ △	$\frac{Ab}{ab}$ ♥	$\frac{aB}{ab}$ ○

Legend:

- : blue, smooth
 △ : yellow, wrinkled
 ♥ : blue, wrinkled
 ○ : yellow, smooth

F₁ phenotypes: $\underbrace{\text{Blue, smooth ; yellow, wrinkled}}_{\text{Parental}} ; \underbrace{\text{blue, wrinkled ; yellow, smooth}}_{\text{Recombinant}}$

[no marks awarded if no stick diagram is used to indicate linkage]

- 1 Parental gametes;
- 2 F₁ genotypes;
- 3 F₁ genotypes correspond to phenotypes / legend for punnett square;
- 4 Correct presentation of genetic diagram (circled gametes, Indication of recombinants and non-recombinant gametes, Indication of recombinants and non-recombinant phenotypes;);

(d) Comment on the locations of these two genes' loci. Show your calculations clearly.

-[2]
- 1 Recombination frequency = No of recombinants/total no of offspring *100;
 - 2 = (355/3312)*100 = 10.7%;
 - 3 Distance between the two genes = 10.7 centimorgan (cM) / map units;
 - 4 On the same chromosome / genes are linked;

(e) Honeybees were also observed to visit the corn plants for pollen. With reference to honeybees, explain how the environment may affect the phenotype.

-[3]
- 1 Queen bees and worker bees are developed from fertilised diploid eggs
/ have same amount of genetic material;
 - 2 They are phenotypically different
/ worker bees are sterile while the queen bee is fertile
/ worker bees are smaller and have larger mouthparts and modified legs as compared to the queen bee;
 - 3 Difference in phenotype is due to the diet of larva;
(in the first 3 days of hatching, all female larvae feed on royal jelly)
 - 4 after 3 days of hatching, larvae destined to be worker bees are fed with pollen;
 - 5 larvae destined to be queen bees are fed with royal jelly;
 - 6 which stimulates maturation of female reproductive system;

[Q5 Total: 13]

6. Palisade cells are plant cells found within the mesophyll in leaves of many plants, right below the upper epidermis and cuticle. They have both chloroplasts and mitochondria. Exchanges between a mitochondrion, a chloroplast and the cytoplasm surrounding them are shown in the **Fig. 6.1**

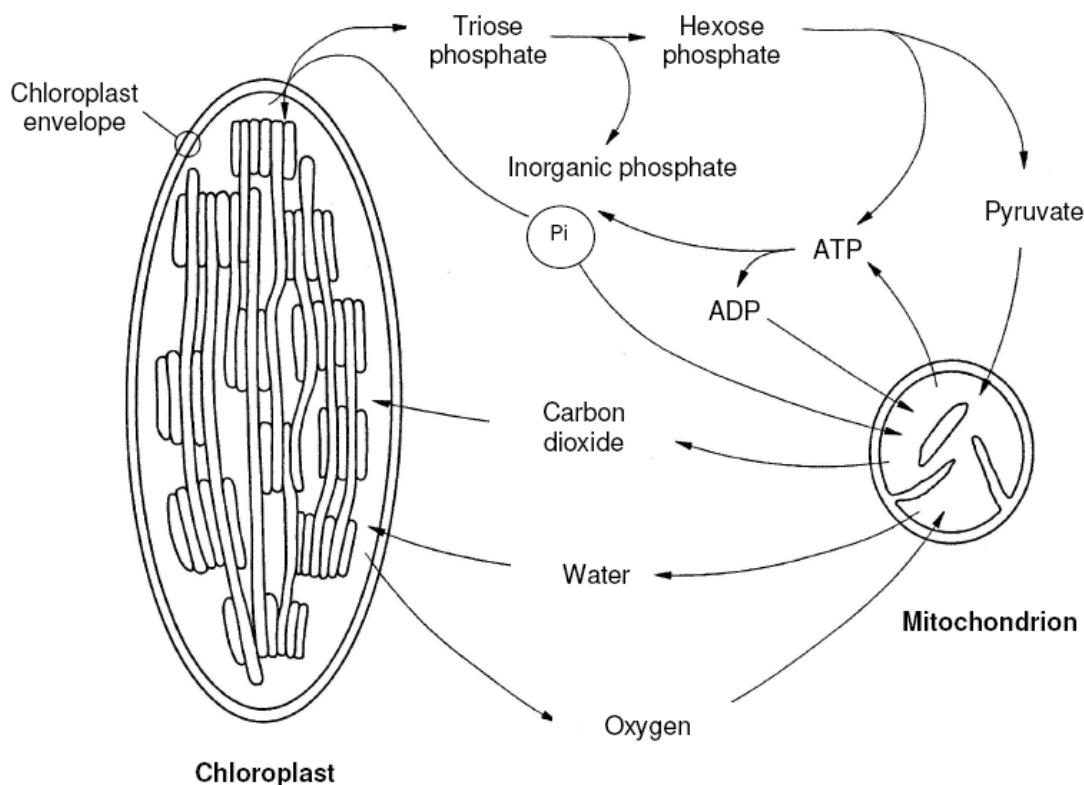


Fig. 6.1

(a) State two structural features of chloroplasts and mitochondria that are also present in prokaryotic cells.

.....[1]

- 1 **DNA** not enclosed by nuclear envelope
- 2 Ribosomes are smaller than those in the cytosol / similar in size to prokaryotic ribosomes / 70S ribosomes
- 3 Circular DNA
- 4 AVP ; e.g. absence of introns in DNA

(b) A leafy shoot is sealed inside a transparent container. The concentration of oxygen in the atmosphere within this container can be measured. In the dark, the oxygen concentration falls. At high light intensities, the oxygen concentration increases. At a particular light intensity, the oxygen concentration in the container remains constant.

With reference to **Fig. 6.1**,

(i) Explain how it is possible for the oxygen concentration to remain constant.

.....[3]

- 1 Oxygen released by chloroplast
- 2 during the photolysis of water
- 3 in the light dependent reaction of photosynthesis

- 4 Oxygen used by the mitochondrion
- 5 as the final electron acceptor in the electron transport chain
- 6 during oxidative phosphorylation

- 7 Compensation point has been reached ;
- 8 where the rate of respiration is equal to the rate of photosynthesis
/ no net change in oxygen concentration;

(ii) Explain why there is no build up in the concentration of inorganic phosphate ions (P_i) inside the mitochondria as a result of the inward passage of P_i .

.....[1½]

- 1 synthesis of ATP from ADP and P_i ;
- 2 in oxidative phosphorylation;
- 3 P_i is also used in the Krebs cycle;

(c) An investigation was carried out to find out the effect of three inhibitors, **A**, **B** and **C**, on the electron transport chain in the mitochondria. In each of three experiments, a different inhibitor was added. The table below shows the state of the electron carriers, **W–Z**, after the addition of inhibitor.

Inhibitor added	Electron carrier			
	W	X	Y	Z
A	oxidised	reduced	reduced	oxidised
B	oxidised	oxidised	reduced	oxidised
C	reduced	reduced	reduced	oxidised

(i) Give the order of the electron carriers in this electron transport chain.

Order [1]
1 Y X W Z

(ii) Explain your answer.

.....[1]
[No marks given if part (i) is wrong]

- 1 presence of oxidized electron carrier imply that the previous carrier has been inhibited
- 2 reduced carriers cannot pass on electrons when inhibited
/ order of carriers linked to sequence of reduction

(d) A student investigated the effects of temperature and light intensity on the rate of photosynthesis of an aquatic plant.

Fig. 6.2 shows the results of the investigation.

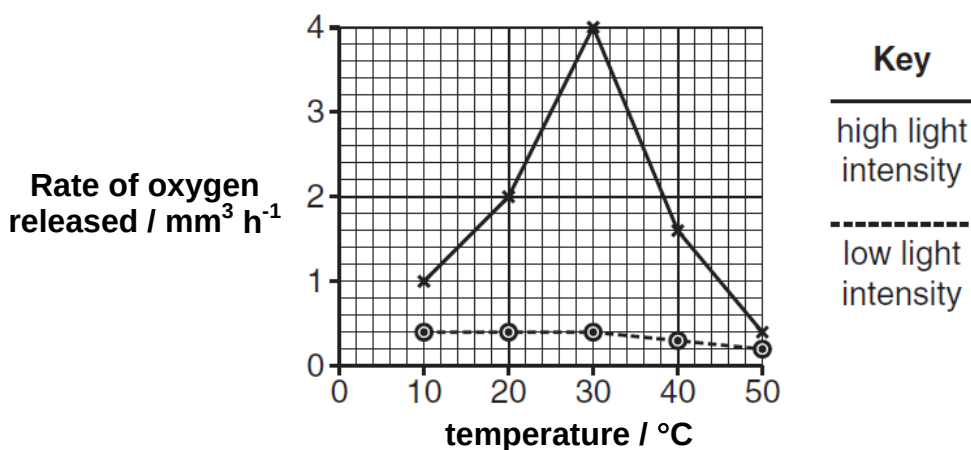


Fig 6.2

With reference to Fig. 6.2,

(i) describe the results of the investigation

.....[2]
in high light intensity

- 1 as temperature increased, the rate of of oxygen released / rate of photosynthesis, increased to a peak and then fell;
- 2 supporting figures (two oxygen values at two different temperatures plus units);

in low light intensity

- 3 as temperature increased, the rate of oxygen released / rate of photosynthesis, remained constant and then fell;
- 4 supporting figures (two oxygen values at two different temperatures plus units);

(ii) suggest explanations for the results for high light intensity above 30°C.

-[1½]
- 1 enzymes denatured / described;
 - 2 so fewer enzyme-substrate complexes / AW;
 - 3 so less photolysis (leads to less oxygen produced);

[Q6 Total: 11]

7. A mammoth can be any species belonging to the extinct genus *Mammuthus*, proboscideans commonly equipped with long, curved tusks. They were members of the family Elephantidae which contains, along with mammoths, the two genera of modern elephants and their ancestors.

About three million years ago, the ancestors of mammoths migrated from Africa into Europe and Asia. There, about 1.7 million years ago, the steppe mammoth (*Mammuthus armeniacus*) evolved and became adapted to the cooler conditions. Then, about 700 000 years ago, as the climate changed and the Arctic became much colder, the woolly mammoth (*Mammuthus primigenius*) evolved. Woolly mammoths showed a number of characteristics that include thick fur, small ears and small tails.

(a) The term *species* is often used in the context of evolution of new species. Explain the meaning of the term *species*.

.....[3]

- 1 Similar morphological, physiological and behavioural features;
- 2 Interbreed in nature to produce viable, fertile offspring;
- 3 Reproductively isolated;
- 4 Occupy same (ecological) niche;
- 5 Shares a common ancestor;
- 6 DNA of organisms of a species nearly identical/have identical gene loci;

(b) Explain how variation and natural selection may have brought about the evolution of the woolly mammoth from the steppe mammoth.

.....[4]

- 1 genetic variations exist within the mammoth population (due to mutation);
- 2 change in selection pressure / much colder condition
- 3 woolly mammoth had a selective advantage in the much colder condition (REJECT "climate change")
- 4 thick fur offers better insulation
- 5 small ears and small tails offers smaller surface area to volume to reduce heat loss
- 6 allowed woolly mammoth to survive till reproductive age
- 7 and pass on their alleles to offspring;
- 8 change in allele frequency of gene pool of population;
- 9 different populations ultimately cannot interbreed to produce viable, fertile offspring (and hence new species formed) / ref. reproductive isolation

(c) A frozen, 43 000 year old woolly mammoth was found in Western Siberia. Its DNA was extracted and sequenced. The phylogenetic relationship of the woolly mammoth, Asian elephant (*Elephas maximus*) and African elephant (*Loxodonta africana*) was then investigated.

(i) Explain how molecular methods can be used to elucidate the evolutionary relationships of the three members of the family Elephantidae.

.....[2]

- 1 Neutral mutations do not confer any selective advantage or disadvantage
- 2 Neutral mutations are accumulated at a relatively constant rate for any particular gene and use as a molecular clock
- 3 Compare DNA sequence of a particular common gene (*or state eg.*) between the three members of the family Elephantidae
- 4 number of differences in genetic sequence is used to calculate the length of time since divergence.

(ii) Explain why molecular evidence is better than morphological evidence in determining phylogeny.

.....[3]

- 1 Quantifiable and open to statistical analysis
- 2 Large quantities of data are required for statistical analysis; however there is little morphological data available
- 3 Unambiguous and objective
- 4 Morphological data may differ depending on the way in which it was classified
- 5 Not affected by convergent evolution / some characteristics may be analogous
- 6 similar morphology may not have been inherited from common ancestor

Additional points

- 7 Neutral mutations are accumulated at a relatively constant rate for any particular gene and use as a molecular clock to date evolutionary events

[Q7 Total: 12]

Section B

Answer **one** question.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

- 8** **(a)** Explain the importance of mitosis and the need for fine control of replication. [6]
- (b)** Outline how the structure of cellulose is related to its specific function. [5]
- (c)** Explain what is meant by primary, secondary, tertiary and quaternary structure of haemoglobin. [9]

[Total: 20]

OR

- 9** **(a)** Describe and explain the transmission of an action potential along a myelinated neurone. [8]
- (b)** Explain how a cholinergic synapse functions. [8]
- (c)** Contrast nervous and hormonal control. [4]

[Total: 20]

2011 SAJC BT2

8(a) Explain the importance of mitosis and the need for fine control of replication. [6]

Max 3

- 1 Mitosis results in genetically identical daughter cells
- 2 To maintain genetic stability of an organism
- 3 Each daughter cell has the same number of chromosomes
- 4 same DNA sequence as the parental cell

- 5 To enable growth and development
- 6 of a **multicellular** organism
- 7 Net increase in the number of cells
- 8 ensures that new cells are identical to existing cells so that they carry out the same function;

- 9 To enable tissue repair / renewal of worn-out or damaged cells;
- 10 Ensures that damaged cells are replaced with exact copies of the original cells in order for the tissue to function properly

- 11 ref. asexual reproduction
- 12 Only one parent / organism is involved in producing new individual organisms
- 13 Ensures that offspring are genetically identical to the parent for continued survival of the species
- 14 retains advantage of the organism in adapting to its environment.

Max 3

Need for fine control of replication

- 1 Semi-conservative DNA replication during S phase in Interphase ;
- 2 Ensure that cells produced are genetically identical ;

- 3 Chromosomes of parent nucleus are already duplicated before mitosis begins ;
- 4 Ensure that 2 sets of identical genome is present before cell division ;

- 5 prevents chromosomal mutation
- 6 ref. aneuploidy or polyploidy would result

- 7 Checkpoint in G1 ensures that cell is large enough with sufficient nutrients before dividing ;
- 8 Checkpoint in M ensures that chromosomes are attached to mitotic spindle so that chromosomes are shared equally between the two daughter nuclei ;
- 9 Checkpoint in G2 assess the success of DNA replication and that the cell is ready for mitosis and cytokinesis.

(b) Outline how the structure of cellulose is related to its specific function. [5]

Great tensile strength:

Structure:

- 1 unbranched polymer
- 2 linked by $\beta(1\rightarrow4)$ glycosidic bonds between β -glucose molecules
- 3 every other glucose monomer is orientated 180° at each successive condensation / alternate units are oriented 180° to each other;
- 4 Each cellulose chain consists of 1,000 or more β -glucose residues.
- 5 The hydroxyl groups (at carbon atom 2) project outwards from each cellulose chain in all directions and
- 6 form hydrogen bonds with neighbouring chains lying parallel to it,
- 7 thus establishing cross-linking between chains.
- 8 Great tensile strength

Function:

- 9 Plant cells are prevented from bursting
- 10 when placed in solutions of higher water potential.

Large intermolecular spaces between macrofibrils:

Structure:

- 11 Large intermolecular spaces
- 12 found between macrofibrils.

Function:

- 13 Allows the passage of water and solute molecules.

(c) Explain what is meant by primary, secondary, tertiary and quaternary structure of haemoglobin. [9]

Primary structure

- 1 Refers to the number and sequence of amino acids
- 2 making up each haemoglobin polypeptide (individual α and β subunits)
- 3 ref (each α -chain is) 141 amino acids long and (each β -chain is) 146 amino acids long
- 4 Peptide bond involved in joining all amino acid monomers together

Secondary structure

- 5 Refers to the folding of the polypeptide into regular structures
- 6 α -helices / coiling of polypeptide chain into a regular helical conformation.
- 7 The carbonyl oxygen (O atom of the C=O) of each peptide bond of one amino acid is
- 8 hydrogen bonded
- 9 to the hydrogen atom on the amino group of another amino acid
- 10 that is 4 amino acids away in the linear sequence.

Tertiary structure

- 11 the folding of the polypeptide chain into its unique 3-dimensional shape;
- 12 ref. globular shape of haemoglobin
- 13 Amino acids far away in primary structure are brought close together (by R group interaction);
- 14 Non-polar/hydrophobic (side chains of) amino acids are buried in the interior;
- 15 Polar and charged/hydrophilic (side chains of) amino acids are on the surface;
- 16 Bonds involved include hydrophobic interactions, hydrogen bonds and ionic bonds
- 17 between R groups of amino acids in each polypeptide chain

Quaternary structure

- 18 Refers to the arrangement of the polypeptide subunits within a protein that is made up of more than one polypeptide chain
/ spatial arrangement of more than one polypeptide chain
- 19 ref. to the association of 2 α and 2 β subunits to form functional haemoglobin molecule
- 20 Bonds involved include hydrophobic interactions, hydrogen bonds and ionic bonds
- 21 between R groups of amino acids in the four subunits

9(a) Describe and explain the transmission of an action potential along a myelinated neurone. [8]

- 1 ref. resting potential in axon at -70 mV
/ high internal K^+ conc. and high external Na^+ conc
/ overall positive charge outside compared to inside of axon;
- 2 arrival of stimulus cause the opening of some voltage-gated Na^+ ion channels,
diffusion of Na^+ ions into the neurone
- 3 results in depolarization;
- 4 when depolarization exceeds threshold potential, all voltage-gated Na^+ ion
channels open
- 5 further diffusion of Na^+ ions in results in an action potential when potential
difference changes to +40 mV;
- 6 at the peak of the action potential, voltage-gated Na^+ ion channels close
- 7 voltage-gated K^+ ion channels open, diffusion of K^+ ions out of axon
- 8 results in membrane repolarisation (membrane potential decreases from +40 mV
to resting level of -70 mV);
- 9 ref. hyperpolarization
- 10 due to voltage-gated K^+ ion channels which remain open after the repolarisation
phase
- 11 refractory period before another action potential can occur
- 12 Na^+-K^+ pumps pump 3 Na^+ ions out for every 2 K^+ ions taken in;
- 13 restores the unequal distribution of Na^+ ions and K^+ ions in the intracellular and
extracellular fluids at resting potential; **REJECT** "restores resting potential"
- 14 depolarization spreads to the node of Ranvier just ahead of the action potential
- 15 due to the lateral diffusion of Na^+ ions to neighbouring node of Ranvier
- 16 opening of Na^+ voltage-gated ion channels leading to diffusion of Na^+ ions into
axon at neighbouring node of Ranvier
- 17 voltage-gated ion channels are concentrated in the nodes of Ranvier
- 18 allow for influx and efflux of ions
- 19 action potentials are generated only at the nodes of Ranvier
- 20 allows for transmission of impulses at speeds much faster than in unmyelinated
neurons
/ ref. saltatory conduction

9(b) Explain how a cholinergic synapse functions. [8]

- 1 Arrival of action potential/nerve impulse at the synaptic knob;
- 2 Opening of voltage-gated Ca^{2+} channels
- 3 Diffusion of Ca^{2+} into pre-synaptic neurone (*since pt 10 is underlined*)
- 4 High concentration of intracellular Ca^{2+} stimulates movement of synaptic vesicles containing neurotransmitter acetylcholine towards pre-synaptic membrane
- 5 Vesicles fuse with the pre-synaptic membrane
- 6 Release of acetylcholine into cleft (via exocytosis)

- 7 Acetylcholine diffuses across synaptic cleft
- 8 Binds to specific cholinergic receptors of ligand-gated Na^+ ion channels embedded on the post-synaptic membrane
- 9 Ligand-gated Na^+ channels open
- 10 Diffusion of Na^+ into post-synaptic neurone

- 11 Post-synaptic membrane becomes depolarised
- 12 Excitatory post-synaptic potential (EPSP) generated
- 13 ref. summation of EPSPs
- 14 Membrane potential reaches threshold potential
- 15 Action potential generated in post-synaptic membrane to initiate an impulse

- 16 These ion channels close and terminates the synaptic response as the neurotransmitters acetylcholine are released from the receptors
- 17 Acetylcholine quickly degraded by enzyme acetylcholinesterase/cholinesterase to acetate and choline
- 18 Choline is reabsorbed into the synaptic knob to be resynthesised into acetylcholine
- 19 enzyme choline acetyltransferase which combines choline with acetyl coA.
- 20 Acetylcholine is then packaged in synaptic vesicles to be re-used

9(c) Contrast nervous and hormonal control. [4]

Differences

	Nervous System	Hormonal System
Complexity	1 More structurally complex (CNS, PNS); / Can integrate vast amounts of information and stimulate a wide range of response;	2 Less structurally complex (endocrine organs/tissues)
Structure & mode of transmission	3 Transmission via a system of neurones that branch throughout the body	4 Transmission by the endocrine glands that secrete hormones into the bloodstream where they are carried to the target organ
Nature of information	5 Neurones conduct electrical signals & chemical signals across synapses	6 Hormones circulate throughout the body via the bloodstream
Localisation of effect	7 Signals transmitted to specific sites by neurone thus response is very localized by allowing fine pinpoint control	8 Hormones secreted into bloodstream is carried to all parts of the body; / Response may be widespread where different target organs may respond to a hormone. E.g. insulin
Effect of Signals	9 Response is usually <u>short-lived</u> (e.g. knee jerk)	10 Response can be <u>long-lasting</u> (e.g. growth)
Response to Time	11 <u>Fast</u> transmission of nerve impulse (up to 100m/s)	12 <u>Slower</u> . May take minutes, hours or days for hormones to be produced, carried by blood to target organ, and for response to occur